EXPERIMENT NO:01

CODE:

```
import matplotlib.pyplot as plt

# Training data
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

# Initial hypothesis
h = ['0', '0', '0', '0', '0', '0']

# Store hypothesis changes
steps = []
scores = []

# FIND-S algorithm
for example in data:
    if example[-1] == 'Yes':
        for i in range(len(h)):
            if h[i] == '0':
                h[i] = example[i]
            elif h[i] != example[i]:
                h[i] = '?'

    steps.append(len(steps) + 1)
    scores.append(sum(1 for x in h if x != '?' and x != '0'))

# Display final hypothesis
print("Final Most Specific Hypothesis:")
print(h)

# Matplotlib visualization
plt.figure(figsize=(8, 5))
plt.plot(steps, scores, marker='o')
plt.xlabel("Training Example")
plt.ylabel("Specific Attributes")
plt.title("FIND-S Algorithm - Hypothesis Generalization")
plt.grid(True)
plt.show()
```

OUTPUT:

```
Final Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

